

Team Briefing - Meeting with Professor David Hughes

Who is David

@all

Professor David Hughes is a **Professor of Personality and Organisational Psychology** based at or affiliated with the Alliance Manchester Business School (AMBS). His career sits at the intersection of psychometric theory and applied organisational practice, meaning he is not purely an academic, but rather has understanding of the real-world constraints of deploying assessment tools in high-stakes environments.

His research covers **the development and critical evaluation of psychometric instruments used in selection contexts.**

He is perhaps most notable for establishing **the Accuracy and Appropriateness validation model**, a framework for assessing whether psychometric tools actually measure what they claim to measure, and whether what they measure actually matters in the context they are deployed.

He has built psychometric tools for genuinely high-risk fields including nuclear energy and law, which tells you something important about his standards: he does not have tolerance for instruments that sound plausible but lack rigorous grounding.

Stanford-Elsevier places him in the top 2% of scientists globally by research impact, which is a meaningful signal of how seriously he is regarded in his field.

He has published directly on the predictive validity of personality questionnaires in selection, which means he has thought carefully and critically about whether the things hiring tools measure actually predict job performance. That is precisely the question Basanite is trying to answer for its own framework.

He has also cc'd his PhD student Claire, who researches Adaptive Personality Regulation, specifically how people adaptively shift their behaviour across tasks to facilitate performance. She is worth engaging seriously during the meeting, not as a courtesy, but because her work is directly relevant to some of Basanite's evaluation dimensions around metacognitive adaptability.

Why He Is Important for Basanite

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First, he is a credibility asset. A professor of his standing engaging seriously with the framework, even critically, is a meaningful signal to the market.

Second, he is a genuine intellectual resource. Hughes has spent decades thinking about exactly these problems in different contexts. A single honest conversation with him is worth months of internal deliberation.

Third, he represents a potential long-term relationship that could anchor the academic legitimacy of the product as it develops. Hughes is the kind of person who, if he believes in the rigour of your approach, could become a genuine collaborator rather than just a reference point.

Meeting Plan

- Introduction of team - Aditya
- Introduction to Basanite - Aditya or Drew
- David sharing his initial thoughts
- Follow up questions - Lynn
- Discussion of collaboration (Optional!) - Aditya

How to Introduce Basanite

@ Drew & Aditya

Hughes already knows the basics from the email. He knows you are building an AI interviewer, that you care about metacognitive dimensions, and that you are at concept validation stage, so the introduction does not need to start from zero.

What it does need to do is four things in quick succession:

- Give him enough substance to actually critique,
- Demonstrate that you have thought rigorously about the problem rather than just built a product
- Translate Basanite's framework into his language without being sycophantic about it
- Open the door for him to push back immediately.

The whole introduction should take no more than five to seven minutes of speaking.

Do not use buzzwords that you use in pitch decks! Do not name-drop investors, valuations, or commercial traction, he is not interested in those signals and they will make the tone feel like a pitch rather than an intellectual conversation.

Do not lead too much with the anti-cheating features, even though they are technically interesting. Starting there may signal that your primary concern is about gaming the system rather than about genuine measurement quality, which is the wrong foot to start on with someone whose career is about measurement rigour.

Do not over-explain the technical architecture of the AI system. Hughes's expertise is in the psychological validity of evaluation instruments, not in AI systems design. What he cares about is what you are trying to measure and whether you have good reason to believe you are measuring it. The mechanism is secondary to that question.

And do not apologise for being a startup or for the early stage of development. You acknowledged the concept validation stage clearly in the email, and he agreed to meet anyway. Own where you are with confidence, and frame the uncertainty as the intellectually interesting part rather than a weakness to excuse.

Suggested Script

"Thank you for making the time, we genuinely appreciate it, and we want to make sure this is useful for you as well as for us. Let me give you a clear picture of what we are building and where we are, and then I would rather spend most of this conversation hearing your thoughts about our assumptions.

The problem we are trying to solve starts from an observation about what the current technical hiring infrastructure actually selects for. Most of the tools in use today (coding tests, structured behavioural interviews, system design rounds) were designed to be objective and

scalable. And on the surface they are. But what they reliably select for is interview preparedness: the ability to perform a convincing approximation of a good candidate in a set-piece evaluation environment. In a world where AI can help any candidate prepare a flawless answer to almost any standard question, that signal is becoming increasingly cheap to fake and increasingly expensive to distinguish from the real thing.

What those tools are structurally blind to are the qualities that actually drive performance in complex technical work: the things that cannot be retrieved from a preparation guide or generated by an AI assistant on the fly. Things like genuine judgment under conditions of incomplete information. The ability to recognise when a team is solving the wrong problem. The metacognitive self-awareness that distinguishes someone who accumulates experience from someone who actually learns from it. These are the qualities we believe are becoming the primary source of human competitive advantage in technical work, precisely because they are the things AI is not replacing.

So what we are building is an AI interviewer that tries to reach those qualities directly.

The core design principle is progressive excavation. Rather than asking candidates to perform competence by answering questions they have prepared for, the system anchors the entire conversation in the candidate's own real experiences, such as projects they actually worked on, decisions they actually made, trade-offs they actually navigated. It then goes progressively deeper into that material: probing the specific reasoning behind choices, asking what they would have done differently, asking under what conditions their solution fails, asking them to predict where their own judgment is most likely to be wrong. The idea is that real capability has edges that performed capability does not. The further you probe, the clearer the difference becomes.

The system evaluates candidates across a set of what we call metacognitive dimensions. At the moment these include things like judgment under ambiguity, the ability to surface tacit knowledge (which is, the gap between what experts know and what they can articulate), creative problem reframing, and what we are loosely calling the capacity to be genuinely transformed by experience rather than just accumulate it. We believe these dimensions share a structural property that makes them both hard to fake and genuinely predictive of performance in complex adaptive work environments.

Now, and this is really why we wanted to speak with you specifically, we are at the stage where we need someone who has spent serious time thinking about what it actually means for an evaluation instrument to measure what it claims to measure. Your Accuracy and Appropriateness framework maps almost perfectly onto the two questions we are least certain about ourselves.

On the accuracy side: we believe our questioning techniques can elicit genuine capability signals rather than performed ones. But we have not yet established whether our evaluation dimensions are sufficiently well-defined and distinct to actually be reliably captured through a

conversational format.

On the appropriateness side: we have theoretical reasons to believe these dimensions predict performance in technical roles. But the predictive validity evidence at the level of specificity we need is thin, and we know it.

We are not here looking for endorsement of what we have built. What would be genuinely useful is a candid assessment of where our theoretical foundations are most likely to break down, the places where we are making assumptions that seem reasonable from inside the product but would not hold up to serious empirical scrutiny. That is the kind of critical perspective that would actually make the product better, and that is what we cannot get from within the team.

So: that is the honest picture of where we are. Where would you like to start?"

Crucial Questions to be Answered During the Meeting

@Lynn

Whether he thinks the metacognitive and judgment-based dimensions Basanite is trying to measure are sufficiently distinct from one another to support reliable measurement, or whether he sees significant overlap that would undermine construct validity. This is the most technically important question for the framework's credibility.

What a rigorous validity framework for an AI-led dynamic interview would actually look like, given that standard inter-rater reliability approaches do not straightforwardly apply when there is no second rater. This is an open question that Basanite genuinely needs to answer, and he may have a view.

Whether the behavioural and cognitive signals the system tries to capture through conversational AI are the kinds of signals that the existing research literature suggests have real predictive validity for performance in complex technical roles — or whether the research is more ambiguous than Basanite's framework assumes.

Where, in his view, AI-led assessment is most likely to introduce systematic error or bias that would be invisible from inside the system. He will have thought about this, and the answer will be genuinely useful.

Whether he thinks the market (meaning employers, HR professionals, and candidates) is psychologically ready to accept the innovated evaluation as legitimate, and what the evidence from analogous transitions (like the shift to online psychometrics) suggests

about adoption patterns and resistance points.

If the conversation is going well, ask him what he thinks the most important unanswered question in AI-assisted talent assessment is: the one the field most needs someone to investigate rigorously. This opens space for collaboration without you having to be the one to name it first.

How to Explore Collaboration Potential Without Asking for Free Labour

@Aditya

The key principle here is that collaboration should emerge from the conversation rather than be proposed at the start of it. Hughes has agreed to give you his time and a critical perspective. Let that be what the first meeting is. If he is asking questions that suggest he finds the problem interesting, that is the moment to mention that you are thinking about how to pursue the validation work more rigorously, and that you would value his perspective on what that might look like.

A few concrete framings that work well in academic-founder conversations:

Frame any potential collaboration around a shared research question rather than around Basanite's commercial needs. The question of whether dynamic conversational AI can produce reliable and valid assessments of metacognitive capability is a genuinely interesting research problem. That is what you are inviting him to work on — not to provide validation services for your startup.

If it feels appropriate, mention that you are thinking about the academic partnership dimension of the product and that you are looking for the right institutional home for some of the validation research. This signals that there is real work to be done and that you are thinking about how to resource it properly, without asking him to commit to anything on the spot.

Be transparent about what Basanite is in a position to offer. Academic collaborations that work usually involve some combination of access to real-world data, co-authorship on research outputs, and in some cases funded research agreements. You do not need to put all of this on the table in the first meeting, but you should not let him think you are expecting his intellectual contribution for nothing. If he asks directly what you are looking for, be honest: you would like to find a way to pursue the validity questions together in a way that creates genuine academic value for him and credibility for the product.

Above all, respect his time. Send a thank-you email within 24 hours of the meeting that captures the key points of the conversation and any follow-up you committed to. If he raised a challenge you do not have an answer to, acknowledge it and tell him you are thinking about it. Researchers value intellectual honesty more than confident closure.